

Newton Interpolating Polynomials

Example 04-01a: Approximate $f(0.45)$ from the following data using Newton polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728

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$$\begin{aligned} f(0.45) \approx p_1(0.45) &= 0.0000000 + 1.7199611 \cdot (0.45 - 0.0) \\ &= 0.7739825 \end{aligned}$$

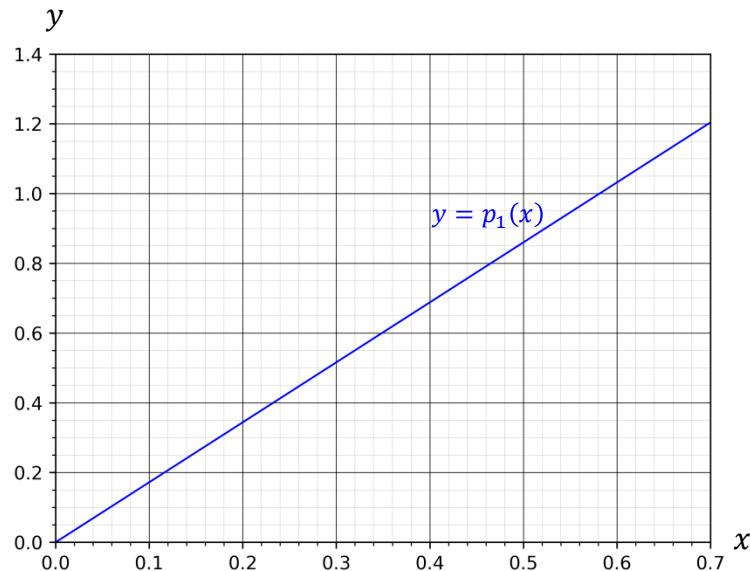
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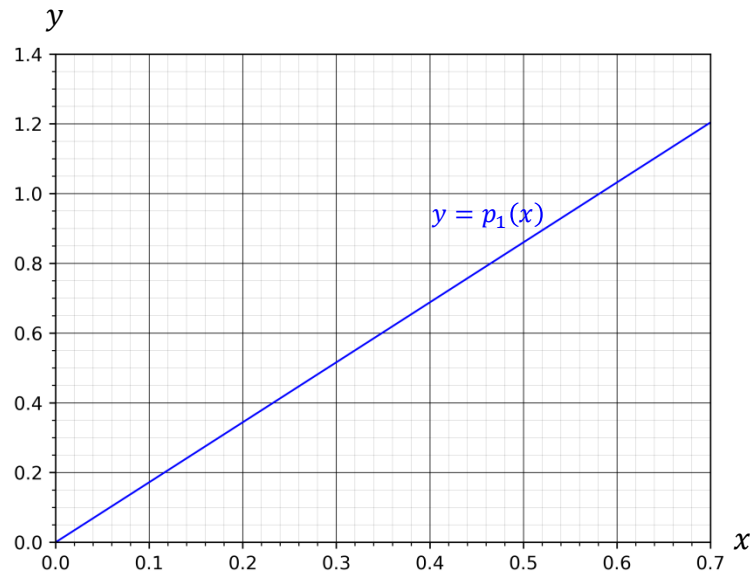
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$$f(x) = -\ln(1 - x)$$



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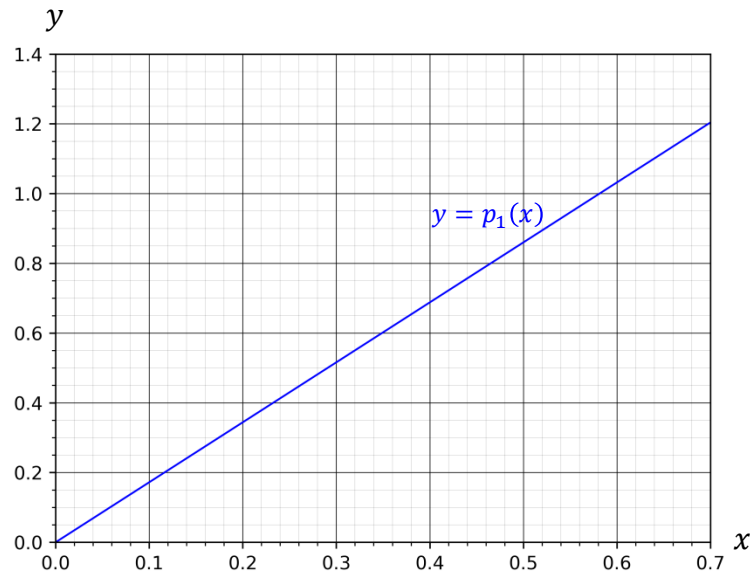
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$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$



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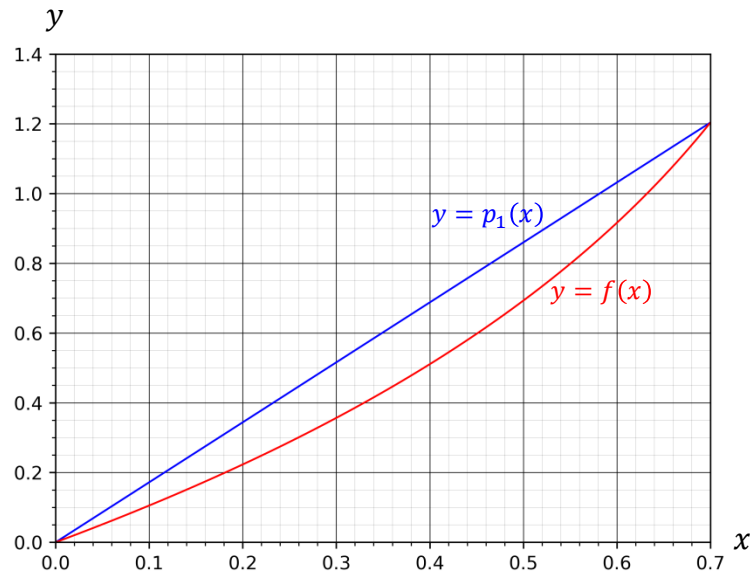
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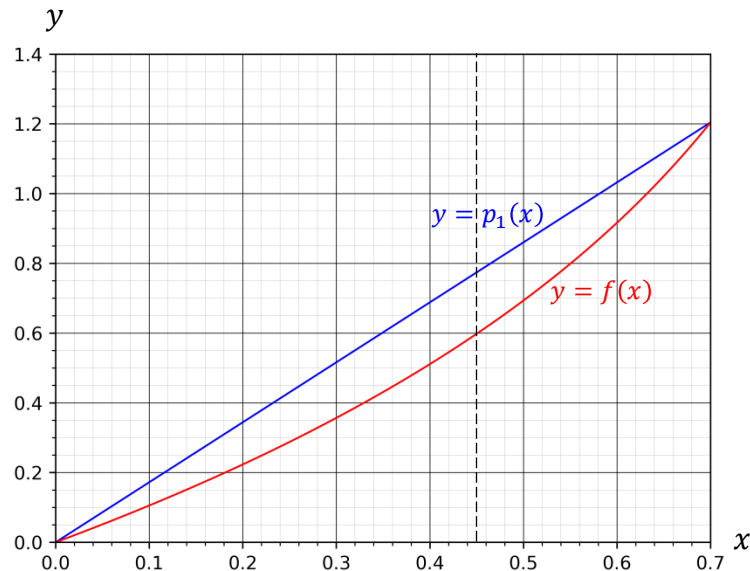
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Newton Interpolating Polynomials

Example 04-01b: Approximate $f(0.45)$ from the following data using Newton polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

Newton Interpolating Polynomials

Example 04-01b: Approximate $f(0.45)$ from the following data using Newton polynomials.

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$$p_2(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1)$$

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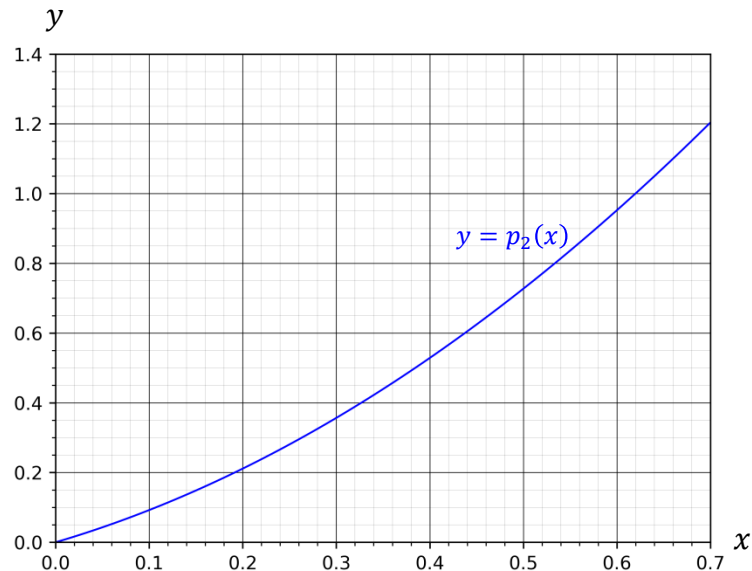
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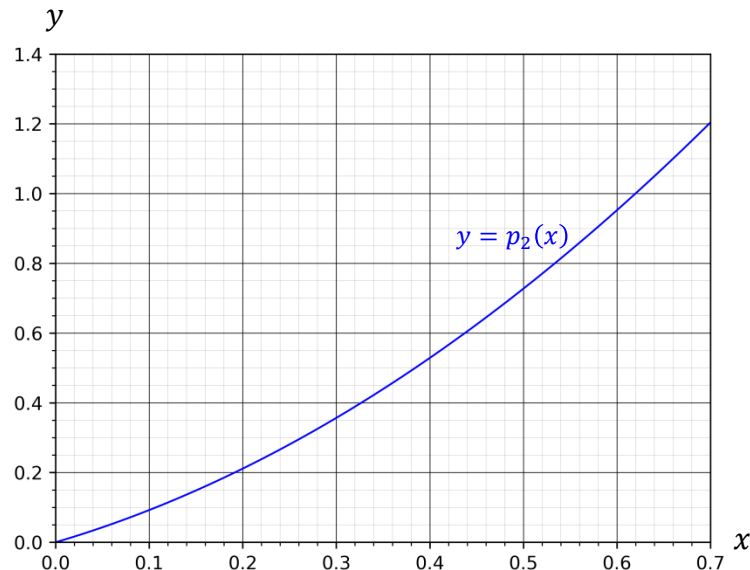
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$$f(x) = -\ln(1 - x)$$

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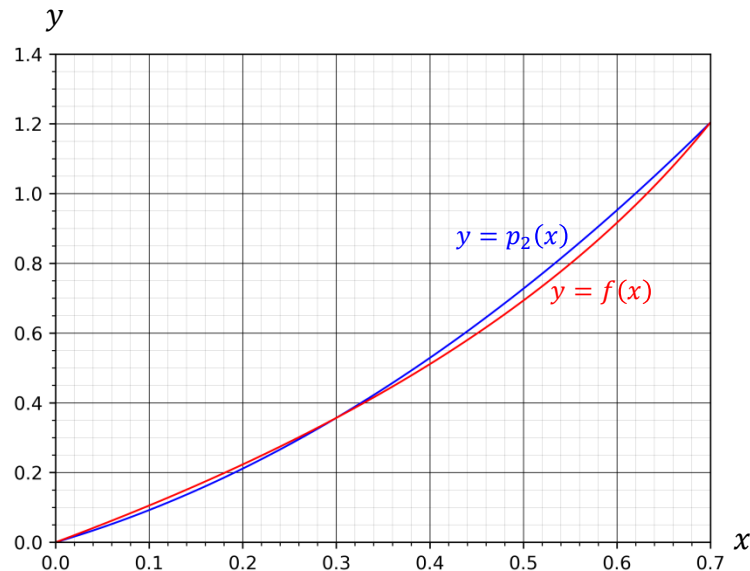
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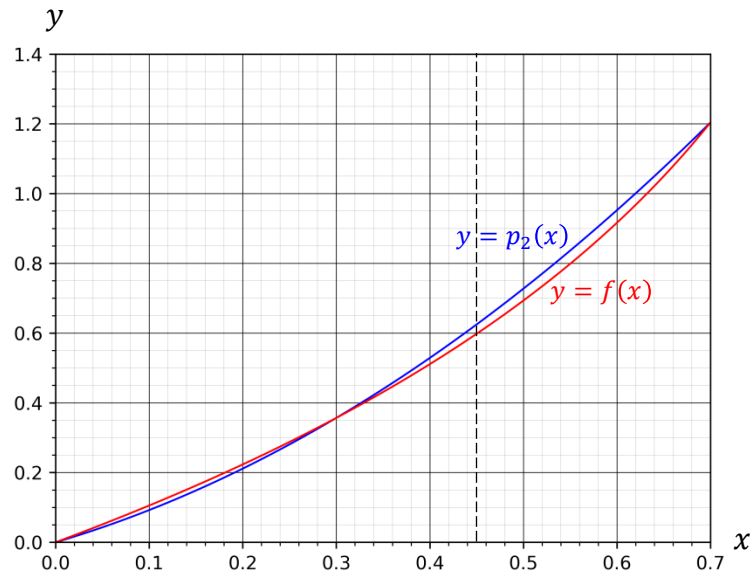
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 &= 0.0000000 + 1.7199611 \cdot (x - 0.0) + 1.3276120 \cdot (x - x_0) \cdot (x - x_1)
 \end{aligned}$$

$$\begin{aligned}
 f(0.45) \approx p_2(0.45) &= 0.0000000 + 1.7199611 \cdot (0.45 - 0.0) \\
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 \end{aligned}$$

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Newton Interpolating Polynomials

Example 04-01c: Approximate $f(0.45)$ from the following data using Newton polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

Newton Interpolating Polynomials

Example 04-01c: Approximate $f(0.45)$ from the following data using Newton polynomials.

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0	0.0	0.0000000
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$$p_3(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) \\ + f[x_0, x_1, x_2, x_3] \cdot (x - x_0) \cdot (x - x_1) \cdot (x - x_2)$$

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$$f(0.45) \approx p_3(0.45) = 0.6045237$$

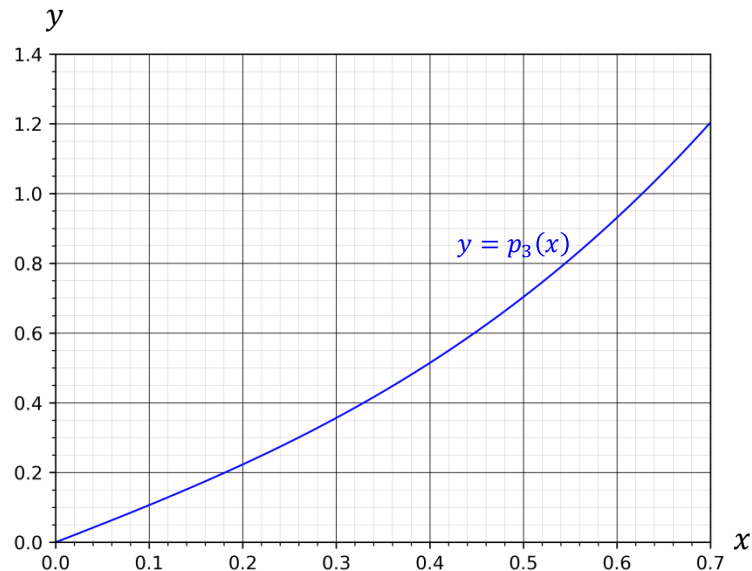
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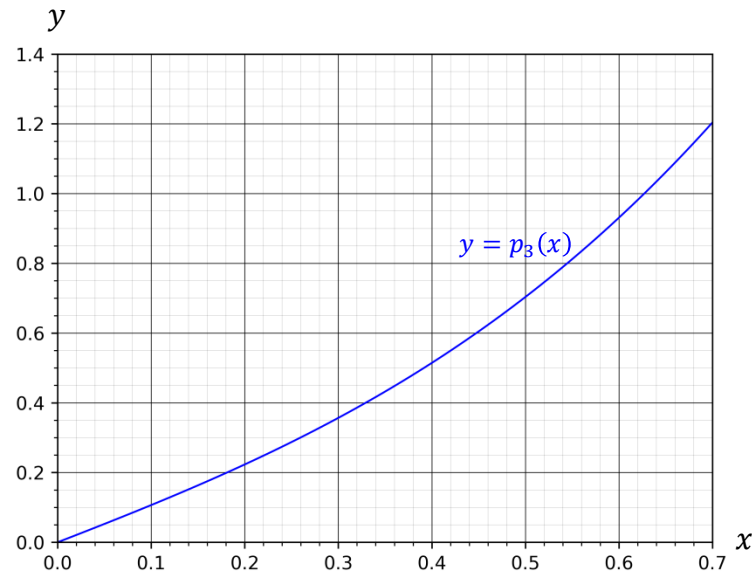
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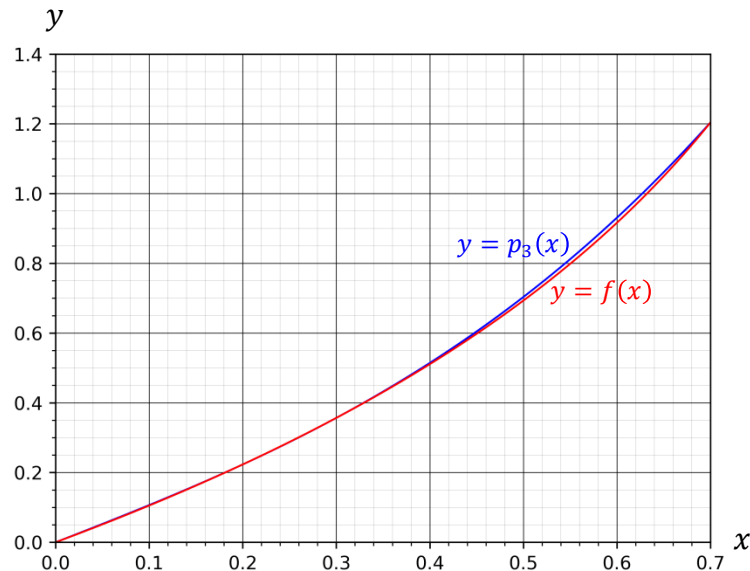
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